

MEDIATEK

MT8676 Android FastRVC User Manual

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Version History

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1 FastRVC

1.1 Overview

This chapter introduces the basic functions of MT8676 FastRVC and how to debug FastRVC problems.

The full name of FastRVC is Fast Rear View Camera. Quick reversing requires the camera feed to be displayed within 4 seconds after a cold boot. Normally, it takes more than ten seconds to display the camera picture after the processes of cameraserver, camerahalservice, surfaceflinger, etc. are started, which cannot meet the requirement of fast reversing; therefore, the FastRVC (instantcam) program was developed to realize the fast reversing function.

1.2 Architecture/Process Overview

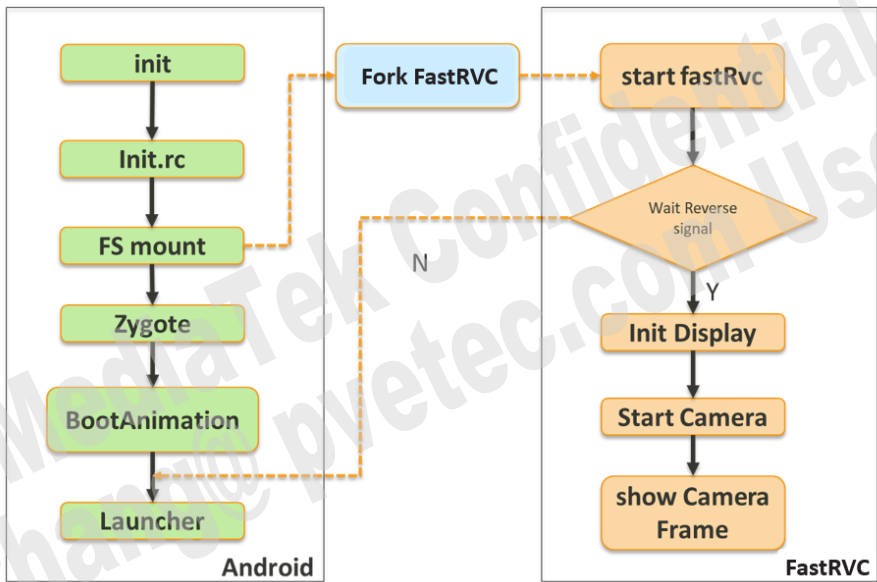


Figure 1-1. FastRVC flow

FastRVC (instantcam) is a process started during the boot phase after the file system is mounted. It mainly monitors reversing events and then sends them to the framebuffer to display the reversing screen.

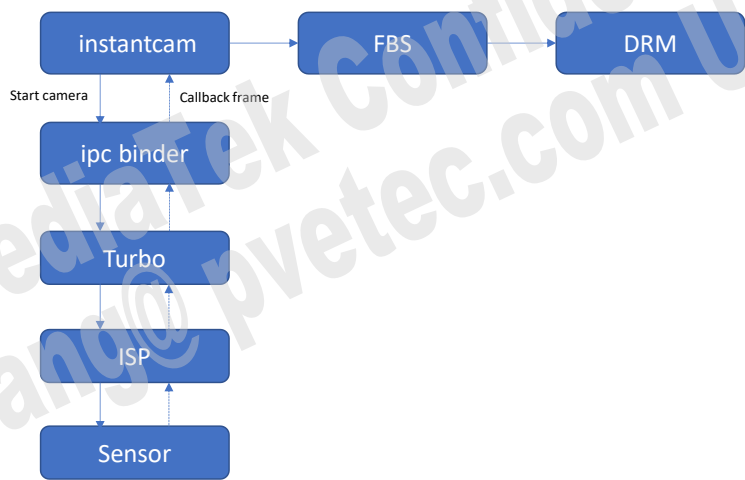


Figure 1-2. instantcam flow

The modules involved in the process of starting FastRVC and displaying the image are shown in Figure 1-2. The functions of each module are as follows:

- IPC binder provides inter-process communication between instantcam and camerahalservice.
- Turbo is a new architecture of MediaTek MW, which is used to manage upper-layer requests and obtain sensor image data from the underlying isp.
- ISP is responsible for processing sensor data and passing it to turbo through conversion.
- FBS is a part of MediaTek display, which is used to process the frame obtained from instantcam and send it to DRM for display.

1.3 Configuration/Customization Guideline

The project enables the fast reversing function, and the configuration is modified as follows:

```
device/mediatekprojects/auto8676p1_64_bsp_fp/device-vext.mk
+ PRODUCT_COPY_FILES +=
$(LOCAL_PATH)/instantcam.rc:$(TARGET_COPY_OUT_VENDOR)/etc/init/instantcam.rc
device/mediatekprojects/auto8676p1_64_bsp_fp/instantcam.rc
+ service instantcam /vendor/bin/instantcam
+ class core
+ oneshot
+ socket rvc_socket stream 660 radio system
+ group audio camera input drmrpc sdcard_r sdcard_rw system media graphics
```

The reverse signal is turned on by default when the machine is powered on:

```
vendor/mediatek/proprietary/hardware/external/mtkInstantCam/instantcam.cpp
-#define MTK_DEFAULT_CAR_REVERSE_SUPPORT 0
+#define MTK_DEFAULT_CAR_REVERSE_SUPPORT 1
```

Specify the sensor type of the current project:

```
vendor/mediatek/proprietary/hardware/external/mtkInstantCam/instantcam.cpp
param.sensorType = Mtk::MTK_SENSOR_FEATURE_SENSOR_TYPE_DMS
```

Indicates using the DMS sensor in muti as the sensor of FastRVC, which can be modified as needed.

1.4 Frequently Asked Questions/Troubleshooting

1.4.1 FastRvc Issue Debug

1.4.1.1 Showing Image Time of FastRvc Quick Reverse Function

Time nodes of the startup phase:

```
adb shell cat /proc/bootprof (unit: ms)
```

There are differences across different hardware platforms, requiring customer debugging. Actual measured data should be used as a reference.

1.4.1.2 FastRVC Customization Function Causing Image Stuck

Customers reported that the screen freezes during the boot phase. The reason is that customers use the `PROP_SET` function in the main thread of the display function, which makes the boot phase more time-consuming. It is recommended that customers avoid setting `PROP_SET` for each frame, and use sub-threads to handle time-consuming operations to avoid jamming the main thread.

1.4.2 FastRVC Debug Log Function

Basic analysis requires providing boot-up phase log, serial port log and logcat log.

For problems with instantcam log that cannot be printed or is missing during the boot-up phase, customers can use the `add_boot_event` function to output the log to bootprof for subsequent customized development. For key log troubleshooting, refer to this method to print and troubleshoot.

1.4.3 Customer Manual Modification Required for Quick Display

FastRVC has some modifications and cannot be uploaded to the main project by default. Therefore, in the version released to customers, this function is not enabled. If customers are sensitive to RVC time, they need to refer to the time optimization part of the MediaTek public version fp project for modification to achieve the goal of fast showing.

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